Exp5

Server

```
#include <stdio.h>
#include <unistd.h>
#include <stdlib.h>
#include <string.h>
#include <arpa/inet.h>
#define PORT 8080
#define MAX 20
int main()
{
    int server_fd, new_socket;
    struct sockaddr_in address;
    socklen_t len = sizeof(address);
    int n, matrix[MAX][MAX];
    char result[100];
    server_fd = socket(AF_INET, SOCK_STREAM, 0);
    address.sin_family = AF_INET;
    address.sin_addr.s_addr = INADDR_ANY;
    address.sin_port = htons(PORT);
    bind(server_fd, (struct sockaddr *)&address, sizeof(address));
    listen(server_fd, 4);
    printf("Waiting for client connection.\n");
    new_socket = accept(server_fd, (struct sockaddr *)&address, &len);
    read(new_socket, &n, sizeof(n));
    read(new_socket, matrix, sizeof(matrix));
    int diagonal = 1, lower = 1, upper = 1;
    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
        {
            if (i > j && matrix[i][j] != 0)
                upper = 0;
            if (i < j && matrix[i][j] != 0)
                lower = 0;
            if (i != j && matrix[i][j] != 0)
                diagonal = 0;
        }
    }
```

```
    if (diagonal)
    {
        strcpy(result, "Diagonal matrix");
    }
    else if (upper)
    {
        strcpy(result, "Upper triangle matrix");
    }
    else if (lower)
    {
        strcpy(result, "Lower triangle matrix");
    }
    else
    {
        strcpy(result, "Neither upper nor lower nor diagonal");
    }
    write(new_socket, result, sizeof(result));
    printf("Matrix Type Sent to Client: %s\n", result);
    close(new_socket);
    close(server_fd);
}
```

Client

```
#include <stdio.h>
#include <unistd.h>
#include <stdlib.h>
#include <string.h>
#include <arpa/inet.h>
#define PORT 8080
#define MAX 20
int main()
{
    int socket_fd;
    struct sockaddr_in address;
    int n, matrix[MAX][MAX];
    char result[100];
    socket_fd = socket(AF_INET, SOCK_STREAM, 0);
    address.sin_family = AF_INET;
    address.sin_port = htons(PORT);
    inet_pton(AF_INET, "127.0.0.1", &address.sin_addr);
```

```
    connect(socket_fd, (struct sockaddr *)&address, sizeof(address));
    printf("Enter the order of matrix:");
    scanf("%d", &n);
    srand(time(0));
    printf("Generated matrix.\n");
    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
        {
            matrix[i][j] = rand() % 50 + 1;
            printf("%4d", matrix[i][j]);
        }
        printf("\n");
    }
    write(socket_fd, &n, sizeof(n));
    write(socket_fd, matrix, sizeof(matrix));
    read(socket_fd, result, sizeof(result));
    printf("Matrix received from client:%s\n", result);
    close(socket_fd);
}
```

Server

```
Waiting for client connection.
Matrix Type Sent to Client: Neither upper nor lower nor diagonal
```

Client

```
Enter the order of matrix:4
Generated matrix.
  39  13  39  39
  26   2  20   3
  20  46   3   9
  21   7  41  21
Matrix received from client:Neither upper nor lower nor diagonal
```